



APSCO

LINKU Project

CAMERA SYSTEM WORKING DOCUMENT

"BLENDER SIMULATION REALISM VALIDATION FOR TETHER
IMAGING"

September 14, 2026

Blender Simulation Realism Validation for Tether Imaging

Prepared by:

¹ Pontificia Universidad Catolica del Peru (PUCP)

Author(s): Jose Sifuentes¹

Status: Working draft, experiments pending

Issue: 1

Revision: 0



Abstract

This working document establishes the methodology to evaluate how realistically Blender (Cycles renderer) can reproduce the visual conditions relevant to LINKU tether imaging, and to what extent that realism supports the use of Blender-rendered images as a stand-in for imagery that cannot yet be captured physically, namely the tether observed under orbital illumination in vacuum.

The validation strategy follows the precedent set by SISPO (Space Imaging Simulator for Proximity Operations), a peer-reviewed Blender-Cycles-based simulator validated by direct pixel-level comparison against real spacecraft camera imagery. The same principle is applied here at a smaller scale: a Blender scene replicating the laboratory dark enclosure and tether proxy is rendered and compared against real photographs taken in that same enclosure, using the project's own reconstruction pipeline (segmentation, endpoint detection, path tracking) as the comparison instrument rather than only visual inspection. Establishing agreement at this laboratory scale, where a real reference photograph exists, is the necessary first step before trusting a second Blender scene configured for orbital illumination, for which no real reference photograph can exist yet.

This document is a working draft. The literature review and methodology are complete; the rendering, capture, and comparison experiments are pending and are marked accordingly throughout the text.

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Introduction

The LINKU stereo vision system has been developed and validated using a laboratory dark enclosure that reproduces the low-texture, high-contrast optical conditions expected during orbital tether observation, as described in the Camera System Report. That validation is limited to what can physically be built and photographed on the ground. It cannot directly test the actual orbital environment: vacuum, unfiltered solar illumination, and the absence of any enclosing structure.

Blender, and specifically its Cycles physically based renderer, offers a way to extend testing beyond the laboratory by generating synthetic images of the tether under conditions that are difficult or impossible to reproduce physically, most importantly orbital solar illumination. However, a synthetic image is only useful as evidence if there is a documented basis for trusting that it behaves like a real photograph would, at least with respect to the properties the reconstruction pipeline actually depends on: radiometric contrast, edge sharpness, and background noise characteristics.

This document defines how that trust is established. The approach does not attempt to prove that a Blender render "looks like" a photograph in a general sense. Instead, it proposes a concrete, two-step validation:

1. Build a Blender scene that replicates the laboratory dark enclosure and tether proxy as closely as possible, render it, and compare the render against a real photograph of the same physical configuration, using the project's own reconstruction pipeline as the comparison instrument rather than visual inspection alone.
2. Only after that laboratory-scale agreement is demonstrated, reconfigure the same Blender scene to represent orbital illumination (vacuum, AM0 solar spectrum, no enclosure) and treat its output as a evidence-supported extrapolation, not an unverified assumption.

This structure follows the precedent of SISPO, a peer-reviewed Blender-Cycles-based space imaging simulator that established its own credibility by comparing renders directly against real spacecraft camera images [1]. The tether observation problem addressed here is smaller in scope, but the underlying validation logic is the same: a simulator is only as trustworthy as the evidence that it matches reality where reality can still be checked.

This document is organized as follows:

- Chapter 1 reviews the relevant literature on physically based rendering realism and on the characterization of solar illumination for space applications.
- Chapter 2 defines the validation methodology, including the two-tier laboratory-then-orbital strategy and the metrics used for comparison.
- Chapter 3 defines the experimental setup required to execute the methodology.
- Chapter 4 defines how results will be evaluated and reported once the experiments are executed.

This document is a working draft. The literature review (Chapter 1) and methodology (Chapter 2) are complete. The experimental setup (Chapter 3) and results (Chapter 4) are pending execution of the Blender scene construction and the laboratory photo capture session.

Chapter 1

Background and Literature Review

1.1 Physically based rendering realism

Blender's Cycles engine is a path tracing renderer that simulates the physical transport of light energy through a scene, including scattering, reflection, and refraction, rather than approximating these effects with non-physical shading tricks. Because of this, Cycles renders can be difficult to distinguish from real photographs when scene geometry, materials, and illumination are defined with physically accurate values rather than visually adjusted ones [2].

The relevant precedent for this document is SISPO (Space Imaging Simulator for Proximity Operations), a Blender-Cycles-based simulator developed for small solar system body fly-by and terrestrial planet surface mission simulation [1]. SISPO is directly relevant for two reasons:

- It validates realism the same way this document proposes to: by comparing rendered output directly against real spacecraft camera imagery, in that case the AMICA instrument images of asteroid Itokawa taken by the Hayabusa spacecraft, rather than by subjective visual inspection.
- Beyond scene lighting and geometry, it models camera-level realism explicitly: dark-current sensor noise drawn from a folded normal distribution, tangential and sagittal astigmatism and comatic aberration through dedicated point-spread functions, and lens distortion through a Brown-Conrady radial and tangential distortion model.

This indicates that scene-level realism (correct geometry and lighting) and camera-level realism (sensor noise and optical aberrations) are treated as two separate concerns that both need to be addressed for a render to be trustworthy as a proxy for a real photograph. Section 2.1 defines which of these two concerns this document addresses first, and which are deferred as future refinements.

1.2 Characterizing solar illumination

A physically based renderer only produces a physically meaningful image if its light sources are configured with real radiometric and spectral values, not with values chosen to "look right" on a monitor. Two properties of the sun matter for this purpose: its spectral color, and its irradiance.

1.2.1 Spectral color

The integral of the measured extraterrestrial solar spectrum matches the integral of a blackbody spectrum at 5772 K [3]. This is the specific figure to use, rather than other round numbers sometimes seen in informal sources, since it is the value stated directly in the SOLAR-ISS reference spectrum paper based on SOLAR/SOLSPEC space-based measurements. Blender's Sun light object supports a Blackbody shader node that takes a color temperature directly and produces the

corresponding spectral color, so this value can be used directly rather than approximated with a manually picked color. A broader review of solar spectral characterization methods and their mutual agreement is given by Thuillier et al. [4].

For applications requiring higher spectral fidelity than a blackbody approximation, dedicated reference spectra exist. The SOLAR-ISS reference spectrum, derived from the SOLAR/SOLSPEC instrument aboard the International Space Station, provides measured solar spectral irradiance from 165 to 3000 nm at high spectral resolution [3]. This is a more rigorous reference than a blackbody approximation but is unlikely to be necessary for this validation, since the reconstruction pipeline operates on broadband intensity images rather than spectral data; the blackbody approximation is expected to be sufficient and is recommended as the default.

1.2.2 Irradiance

Blender’s Sun light strength in Cycles is expressed directly in units of irradiance, watts per square meter (W/m^2), which is the same unit used to report solar irradiance [2]. This makes the value directly settable rather than requiring unit conversion.

Two different irradiance values are relevant depending on which condition is being simulated:

- **Ground-level, atmosphere-attenuated illumination:** commonly cited around $1050 \text{ W}/\text{m}^2$ for direct sunlight after atmospheric absorption and scattering. This is the value used in most Blender outdoor-lighting tutorials, because most Blender scenes represent terrestrial, not orbital, conditions.
- **Extraterrestrial illumination (AM0, Air Mass Zero):** the internationally recommended solar constant (IAU, 2015) is approximately $1361 \text{ W}/\text{m}^2$. This is the traditional engineering reference value, used in standards such as ASTM E490 for spacecraft and photovoltaic component testing, and is recommended as the default for this document. A more recent reprocessing of SOLAR/SOLSPEC space-based measurements reports a total solar irradiance of $1372.3 \pm 16.9 \text{ W}/\text{m}^2$, stated explicitly as $11 \text{ W}/\text{m}^2$ above the IAU 2015 value [3]. The two values differ by less than 1%, and either is defensible; this document adopts $1361 \text{ W}/\text{m}^2$ because it is the more widely referenced engineering value, and notes $1372.3 \text{ W}/\text{m}^2$ as an available refinement if higher precision is later judged necessary.

The orbital-condition Blender scene described in this document must use one of the two AM0 values above, not the ground-level value. This distinction is easy to overlook because most publicly available Blender tutorials and default scene templates assume a terrestrial, atmosphere-filtered Sun.

The ASTM E490 standard itself has not been directly reviewed for this document; the $1361 \text{ W}/\text{m}^2$ figure above is sourced from its role as a comparison point in Meftah et al. [3], not from a direct reading of the standard. Obtain and verify against ASTM E490 directly if this figure needs to be cited as a primary requirement rather than as background justification.

1.3 Validating rendered realism against real images

Beyond space-specific simulators, general computer graphics research has also studied how to determine whether a rendered image is convincingly realistic when compared against real photographs, typically through structured user studies or quantitative image similarity metrics, as in the two-alternative forced-choice study by Kluge and Staadt [5]. This body of work is relevant background but is not the primary methodology adopted here: rather than asking whether a human observer finds the render convincing, Chapter 2 proposes asking whether the project’s own automated reconstruction pipeline produces consistent results on the render and on the real photograph. This is a stricter and more directly relevant test for this application, since the pipeline, not a human viewer, is the actual downstream consumer of the images.

Chapter 2

Validation Methodology

2.1 Scope

This document addresses scene-level realism first: correct geometry, correct tether proxy appearance, and correct illumination (spectral color and irradiance, as defined in Chapter 1). Camera-level realism, sensor noise, and optical aberrations of the specific camera model in use, is treated as a future refinement and is not required for the first validation pass, since the reconstruction pipeline's segmentation and tracking stages are expected to be more sensitive to illumination and contrast than to these secondary effects. This should be revisited if the first validation pass shows disagreement that scene-level factors cannot explain.

2.2 The core problem: no ground truth exists for the orbital condition

The central difficulty is that the condition of real interest, the tether photographed in orbit, cannot be photographed for real. There is no ground-truth image to compare against for that condition, which means realism cannot be validated directly where it matters most.

The proposed solution is to validate indirectly, through a condition that is one step removed from orbit but still has a real photographable reference: the laboratory dark enclosure. Figure 2.1 summarizes this two-tier strategy.

2.3 Comparison instrument: the reconstruction pipeline itself

Rather than relying on visual inspection or generic image-similarity metrics to decide whether the render "looks real enough," this methodology reuses the project's own reconstruction pipeline as the comparison instrument. The logic is direct: the pipeline is the actual downstream consumer of these images in operational use, so agreement at the pipeline's output is a more relevant realism criterion than agreement at the pixel level or to a human observer.

For a matched pair of images, one real and one simulated, of the same tether configuration, the following outputs are compared:

- **Segmentation mask:** coverage percentage and, where a pixel-level correspondence can be established, intersection-over-union between the real and simulated mask.
- **Endpoint detection:** position of the two detected tether ends, compared directly since both images represent the same physical configuration.
- **Path tracing:** coverage percentage and path length from the SmartWireTracker output.

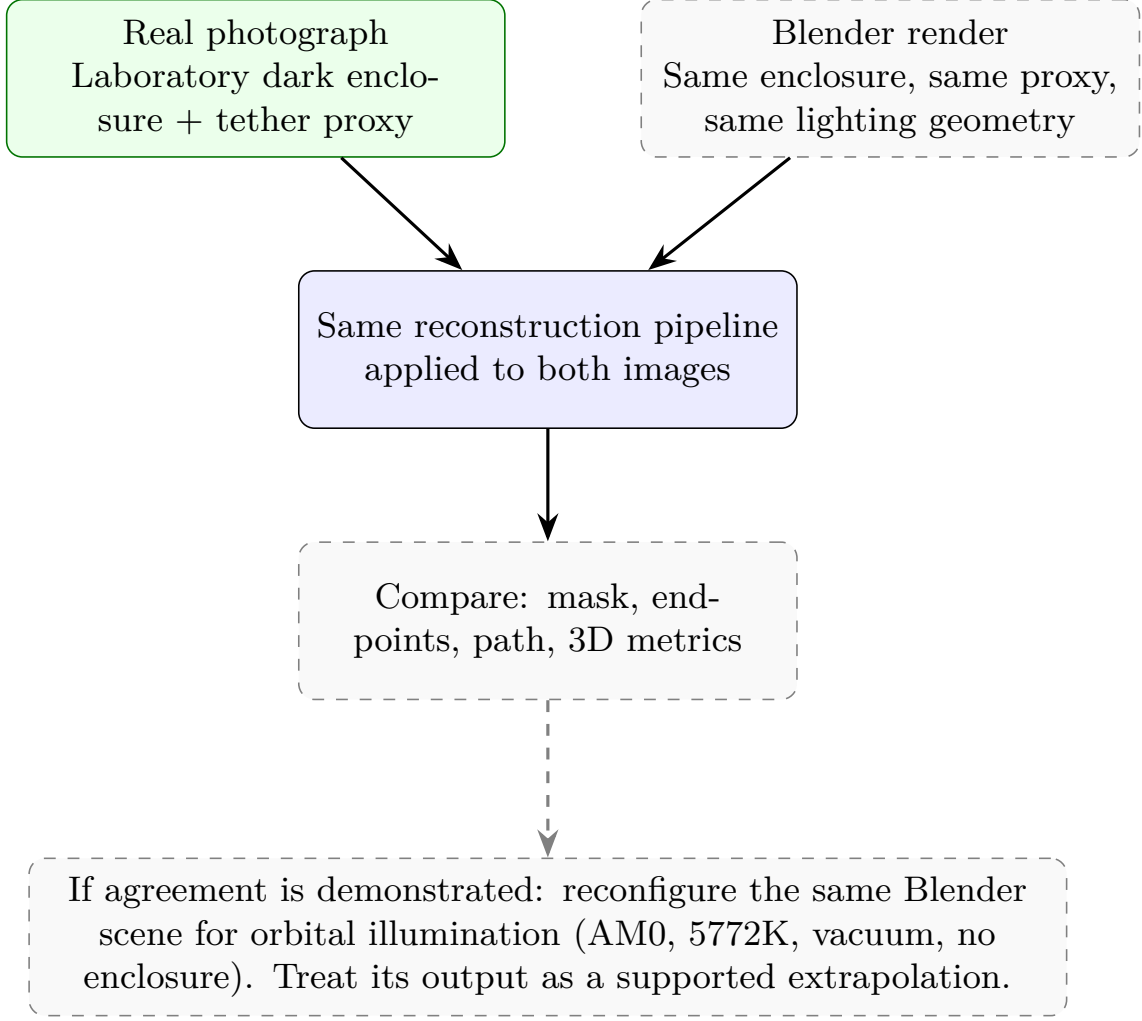


Figure 2.1: Two-tier validation strategy. Solid green boxes have a real reference available today. Dashed gray boxes are pending. The orbital-condition render (bottom) is only trusted to the extent that the laboratory-condition render (top) is shown to agree with its real photographic counterpart.

- **3D reconstruction metrics:** total length, straightness, and curvature, when both images are available as part of a stereo pair.

Close agreement across these outputs is the evidence required before the orbital-condition render is treated as informative rather than speculative.

2.4 Independent variables to control

To keep the comparison meaningful, the following must be matched between the real photograph and the Blender render as closely as physically and digitally possible:

- Tether proxy geometry and pose (same physical configuration, or a measured and replicated one).
- Camera intrinsic parameters (focal length, sensor size, field of view) matching the calibrated real camera.
- Camera extrinsic parameters (position and orientation relative to the tether) matching the real capture setup.
- Illumination geometry (light source position and direction relative to the tether and camera).

- Illumination spectral color and irradiance, per Chapter 1, matched to the real dark-enclosure light source for the laboratory-condition render, and set to AM0/5772K only for the orbital-condition render.

Pending: measure the real dark-enclosure light source's actual spectral color temperature and irradiance (if not already characterized in the Camera System Report) so the laboratory-condition Blender render matches the real light source rather than an assumed value.

Chapter 3

Experimental Setup

This chapter defines what needs to be built and captured to execute the methodology of Chapter 2. All items in this chapter are pending; none have been executed yet.

3.1 Blender scene construction

Pending: build the Blender scene. Required elements: (1) tether proxy geometry matching the real proxy's diameter and material appearance, (2) dark-enclosure interior geometry and matte-black material properties, (3) camera object with intrinsic parameters matching the calibrated real camera, (4) light source configured per Chapter 1: Blackbody shader at approximately 5772K, Sun strength set to the real light source's measured irradiance for the laboratory-condition render, and to 1361 W/m² (AM0) for the orbital-condition render.

3.2 Laboratory photograph capture

A reference stereo pair has been selected from the existing capture archive: the tether enters the frame from above the camera and falls toward the background, chosen because this configuration is simple to reproduce and reason about (no self-crossing, no ambiguous topology) for a first matched-pair test.

- Left image: `data/captures/stereo_20260213_140535/left.jpg`
- Right image: `data/captures/stereo_20260213_140535/right.jpg`
- Metadata: `data/captures/stereo_20260213_140535/metadata.json` (capture resolution 1920×1440, baseline 100 mm, synchronization quality "EXCELENTE", 10.4 ms time difference between cameras)

Pending: this pair was captured before the current calibration was invalidated (Chapter 7 recalibration item, Camera System Report), so it is usable as a 2D reference for mask/endpoint/path comparison but not for 3D reconstruction comparison. Also pending: document the physical pose of the tether proxy for this specific capture (approach angle, distance from camera, support/attachment point) so it can be replicated in the Blender scene.

3.3 Matched-pair generation

Pending: for each real photograph captured above, render the corresponding Blender scene from the same camera position, orientation, and tether pose, producing one matched real/simulated pair per configuration tested. A minimum of 3 to 5 matched pairs, covering different tether poses (straight, curved, and one closer to a self-crossing configuration if feasible), is recommended before drawing conclusions from a single case.

3.4 Orbital-condition render generation

Pending, and dependent on satisfactory agreement in the laboratory-condition comparison (Chapter 4): reconfigure the validated Blender scene to represent orbital conditions (vacuum, no enclosure, AM0 solar illumination, unobstructed star-field or black background) and render the tether under representative deployment geometries.

Chapter 4

Evaluation and Reporting Plan

This chapter defines how results will be evaluated and reported once the experiments of Chapter 3 are executed. No results exist yet; this chapter defines the criteria that will be applied to them.

4.1 Reporting format

For each matched real/simulated pair, the following will be reported side by side:

- The real photograph and the Blender render, for visual reference.
- The segmentation mask obtained from each, with coverage percentage.
- The detected endpoints from each, with pixel-distance between the real and simulated endpoint positions.
- The traced path from each, with coverage percentage and path length.
- A qualitative note on any visible discrepancy and a hypothesis for its cause (for example: material reflectance mismatch, light source position error, or camera intrinsic mismatch).

4.2 Interpretation criteria

Pending decision, to be set before running the experiment rather than after seeing the results, to avoid post-hoc rationalization: define the quantitative agreement threshold (for example, endpoint position error below N pixels, mask coverage within X percentage points) below which the laboratory-condition render is considered validated and the orbital-condition extrapolation is considered supported.

4.3 Outcome paths

Two outcomes are possible once the laboratory-condition comparison is complete:

- **Agreement within the defined threshold:** proceed to generate and use orbital-condition renders, with the explicit caveat that confidence is inherited from laboratory-scale agreement and has not been independently verified for the orbital condition itself.
- **Disagreement beyond the defined threshold:** do not proceed to orbital-condition renders until the source of disagreement is identified and corrected. Likely candidates, in rough order of expected impact, are light source spectral or irradiance mismatch, tether material reflectance mismatch, and camera intrinsic parameter mismatch.

Chapter 5

Conclusions and Status

This document defines a concrete, literature-grounded methodology for establishing whether Blender-rendered images of the LINKU tether can be trusted as a proxy for photographs that cannot yet be taken physically, most importantly the orbital illumination condition. The methodology follows the precedent set by SISPO, validating simulator realism through direct comparison against real reference imagery rather than through visual inspection, and adapts it to this project by using the reconstruction pipeline itself, rather than a human observer, as the comparison instrument.

The literature review identified the correct physical values to use for solar illumination in Blender: a Blackbody shader at approximately 5772K for spectral color, and a Sun strength of approximately 1361 W/m² (AM0, extraterrestrial solar constant) for orbital-condition irradiance, as opposed to the lower ground-level value assumed by default in most Blender tutorials.

5.1 Current status

- Literature review: complete.
- Methodology definition: complete.
- Blender scene construction: pending.
- Laboratory photograph capture: pending.
- Matched-pair comparison: pending.
- Orbital-condition render generation: pending, and dependent on the laboratory-condition result.

5.2 Next steps

1. Measure or confirm the real dark-enclosure light source's spectral and irradiance characteristics.
2. Build the Blender scene replicating the laboratory dark enclosure.
3. Capture the reference real photographs.
4. Render the matched simulated counterparts and run the comparison defined in Chapter 2.
5. Decide, using the interpretation criteria of Chapter 4, whether to proceed to orbital-condition renders.

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